VI SEMESTER (OPEN ELECTIVE-II)	L	T	P	C
	2	-	2	3
20CE6O01 - ENVIRONMENTAL POLLUTION AND CONTROL				

COURSE OUTCOMES:
Students are able to
CO1: Identify the air pollutant causes and control devices. (K2)
CO2: Differentiate the treatment techniques used for sewage and industrial wastewater treatment methods. (K2)
CO3: Understand the fundamentals of solid waste management, practices adopted in his town/village and its importance in keeping the health of the city. (K2)
CO4: Know the causes for noise pollution and ISO14000 standards. (K2)
CO5: Know Treatment and management of hazardous waste. (K2))

UNIT -I : Air Pollution
Air pollution causes-control methods–particulate control devices – methods of controlling Gaseous Emissions – Air quality standards.

UNIT -II: Industrial wastewater Management
Strategies for pollution control – Volume and Strength reduction – Neutralization – Equalization – Proportioning – Common Effluent Treatment Plants – Recirculation of industrial wastes – Effluent standards.

UNIT -III : Solid Waste Management
Solid waste characteristics–basics of on-site handling and collection–separation and processing – Incineration- Composting-Solid waste disposal methods – fundamentals of land filling.

UNIT -IV: Noise Pollution
Noise standards, Measurement and control methods – Reducing residential and industrial noise – ISO14000

UNIT -V: Hazardous Waste
Characterization – Nuclear waste – Biomedical wastes – Electronic wastes – Chemical wastes – Treatment and management of hazardous waste-Disposal and Control methods.

Text books:
1. K. Sasi Kumar, S.A. Gopi Krishna ,"Solid Waste Management",PHI New Delhi,2014.
2. D. Srinivasan, "Environmental Engineering", PHI Learning Private Limited, New Delhi, 2011.

References books:
1. Ruth F. Weiner and Robin Matthews , 'Environmental Engineering', 4th Edition Elesevier, 2003.
2. J.G. Henry and G.W. Heinke,'Environmental Science and Engineering'– Pearson Education,2002
3. Mackenzie L Davis & David A Cornwell, "Environmental Engineering ',McGraw Hill Publishing,2002.
4. Howard S. Peavy, Donald R. Rowe, Teorge George Tchobanoglus,"Environmental Engineering", Mc-Graw-Hill Book Company, New Delhi, 1985.

E-Resources:
1. https://nptel.ac.in/courses/123105001